IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of methodologies

- Collected Falcon 9 launch data from the SpaceX REST API and by scraping Wikipedia
- Cleaned the data and created a landing outcome label (1 = landed, 0 = did not land)
- Explored the data with charts and SQL queries
- Built an interactive Folium map and a Plotly Dash dashboard
- Trained and tuned four classification models to predict landing success

Summary of results

- **EDA:** 66.7% of first stages landed overall. The yearly success rate rose from 0% (2010 to 2013) to about 90% in 2019
- **Interactive analytics:** KSC LC-39A has the highest success rate (76.9%). All sites sit near the coast, a railway and a highway, and away from cities
- **Predictive analysis:** all four models scored 83.3% test accuracy (15 of 18). The Decision Tree had the best cross-validation accuracy (0.891)

Introduction

Project background and context

SpaceX advertises Falcon 9 launches for about 62 million dollars, while other providers charge upwards of 165 million dollars. Much of that saving comes from landing and reusing the first stage.

If we can predict whether the first stage will land, we can estimate the cost of a launch. That is useful to any company that wants to bid against SpaceX for a launch contract.

Questions this project answers

- Which factors (payload mass, orbit, launch site, flight number) are linked to a successful landing?
- Has the landing success rate improved over time?
- Which launch sites have the highest success rate, and where are they located?
- Which classification model best predicts whether the first stage will land?

Section 1

Methodology



Methodology

Executive Summary

- **Data collection methodology:** SpaceX REST API (launch, rocket, launchpad, payload and core endpoints) and web scraping of the Wikipedia Falcon 9 launch table with BeautifulSoup
- **Perform data wrangling:** filtered to Falcon 9 launches, filled missing payload mass with the mean, and turned landing outcomes into a binary Class label
- **Perform exploratory data analysis (EDA) using visualization and SQL:** scatter, bar and line charts with Matplotlib and Seaborn; SQL queries on SQLite
- **Perform interactive visual analytics using Folium and Plotly Dash:** launch site map with outcome markers and distance lines; dashboard with a site dropdown and payload slider
- **Perform predictive analysis using classification models:** standardized the features, split train/test, tuned Logistic Regression, SVM, Decision Tree and KNN with GridSearchCV (10-fold cross-validation), and compared test accuracy

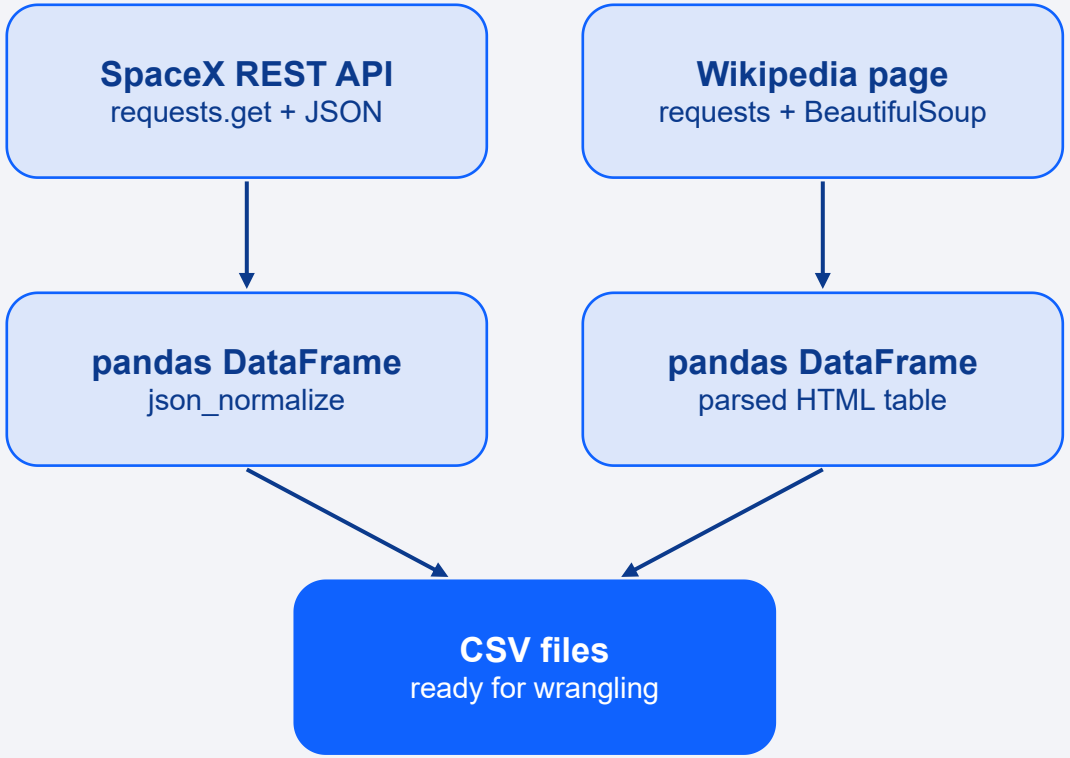
Data Collection

Launch data came from two sources:

- **SpaceX REST API:** structured JSON records of past launches. IDs in each record (rocket, launchpad, payload, core) were used to call further endpoints for booster name, site coordinates, payload mass, orbit and landing details.
- **Wikipedia web scraping:** the "List of Falcon 9 and Falcon Heavy launches" page, parsed with BeautifulSoup to get a second, independent record of launches.

Both were converted to pandas DataFrames and saved as CSV files for the next steps.

The live SpaceX API was offline (HTTP 522) during this project, so the API steps used IBM's archived copies of its data.

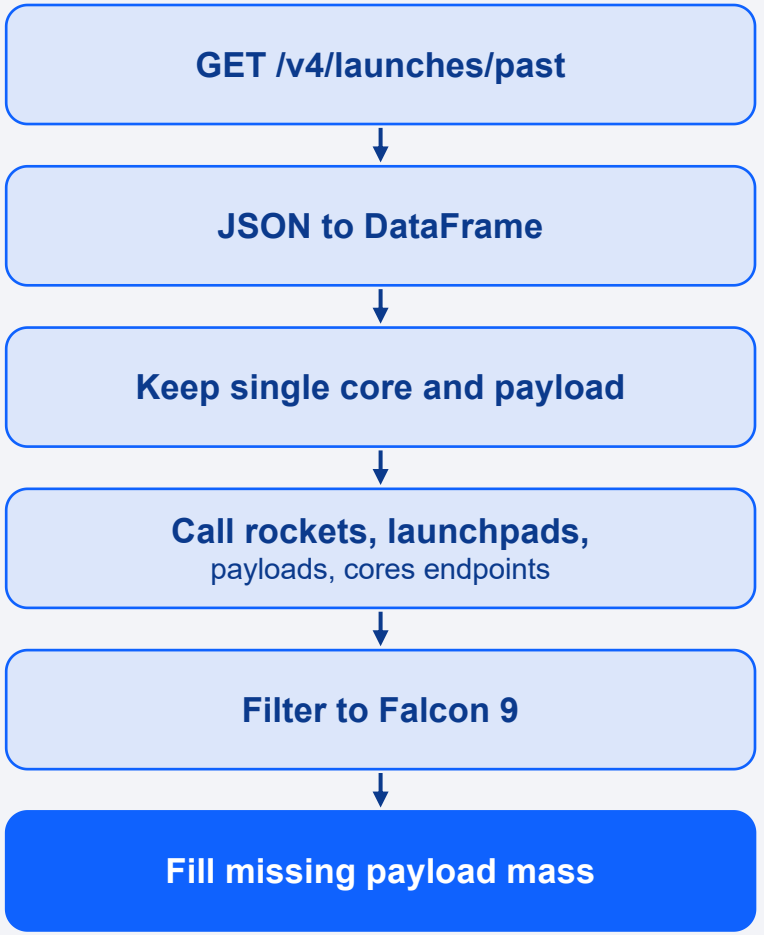


Data Collection – SpaceX API

- Requested past launches from api.spacexdata.com/v4/launches/past
- Flattened the JSON response with pd.json_normalize
- Kept launches with a single core and a single payload
- Used helper functions to call the rockets, launchpads, payloads and cores endpoints by ID
- Filtered out Falcon 1 launches to keep only Falcon 9
- Replaced missing PayloadMass values with the column mean

Note: api.spacexdata.com returned HTTP 522 during this project (the API project was archived in June 2026). Launch records came from IBM's static copy of the API response; the ID lookups could not run, so later labs used IBM's prepared dataset.

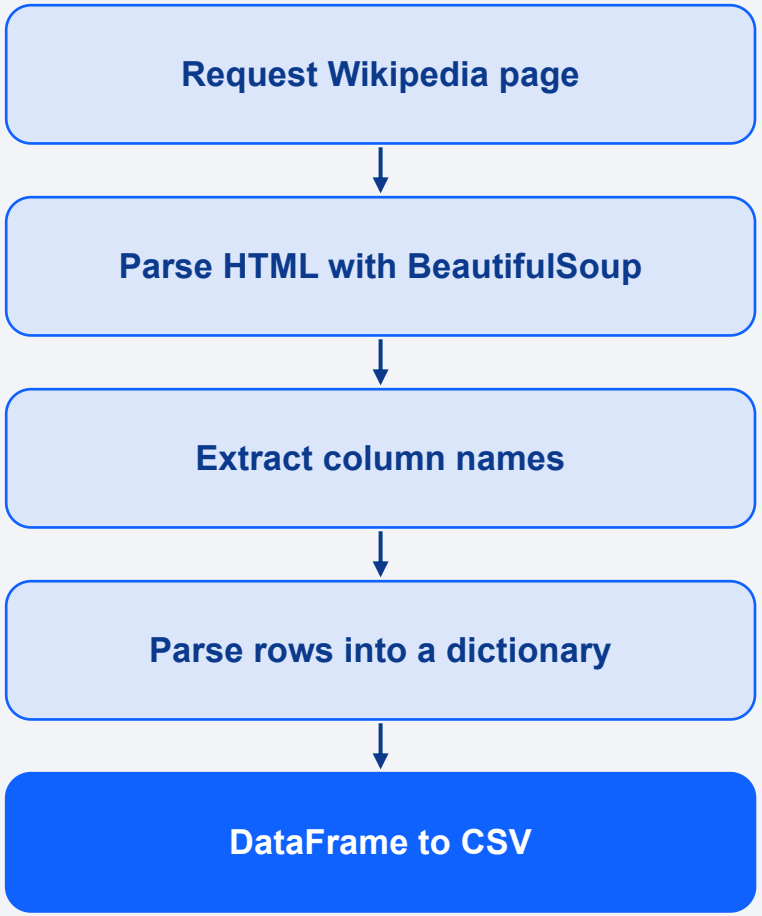
GitHub notebook: jupyter-labs-spacex-data-collection-api.ipynb



Data Collection - Scraping

- Requested the Wikipedia page "List of Falcon 9 and Falcon Heavy launches" with requests
- Created a BeautifulSoup object from the HTML response
- Found all tables and extracted the column names from the launch table header
- Looped through the table rows to fill a dictionary of launch records
- Converted the dictionary to a DataFrame and saved it as CSV

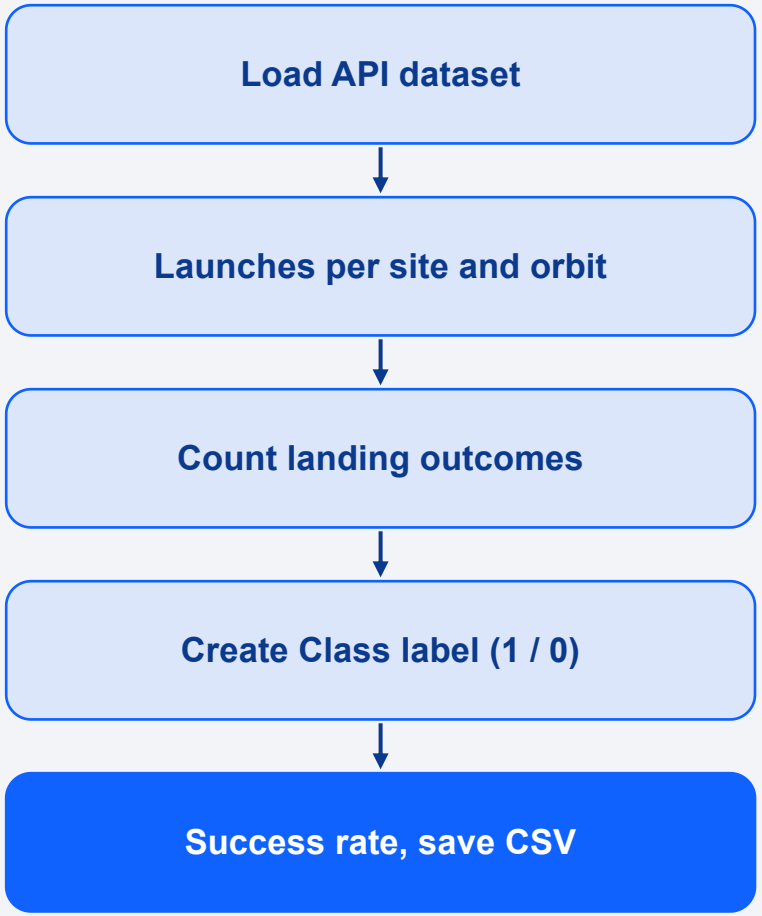
GitHub notebook: jupyter-labs-webscraping.ipynb



Data Wrangling

- Counted launches per launch site and per orbit type
- Counted each landing outcome, for example True ASDS (landed on a drone ship) or False Ocean (failed ocean landing)
- Grouped outcomes into successful and unsuccessful landings
- Created the label column Class: 1 if the first stage landed, 0 if it did not
- Calculated the overall success rate and saved the result as CSV

GitHub notebook: labs-jupyter-spacex-Data_wrangling.ipynb



EDA with Data Visualization

Chart type	What was plotted	Why this chart
Scatter plots	Flight Number vs. Launch Site, Payload vs. Launch Site, Flight Number vs. Orbit, Payload vs. Orbit	Show how two variables relate, colored by landing outcome
Bar chart	Success rate of each orbit type	Compare one value across categories
Line chart	Yearly average success rate	Show the trend over time

The features found useful here were then one-hot encoded for modeling.

GitHub notebook: edadataviz.ipynb

EDA with SQL

- Unique launch site names
- 5 records where the launch site begins with 'CCA'
- Total payload mass carried for NASA (CRS)
- Average payload mass for booster version F9 v1.1
- Date of the first successful ground pad landing

- Boosters that landed on a drone ship with payload between 4000 and 6000 kg
- Total number of successful and failed mission outcomes
- Booster versions that carried the maximum payload mass (subquery)
- Failed drone ship landings in 2015, with booster version and launch site
- Landing outcomes between 2010-06-04 and 2017-03-20, ranked by count

SQL used: SELECT DISTINCT, LIKE, SUM, AVG, MIN, COUNT, GROUP BY, ORDER BY, subqueries, and date filtering on SQLite.

GitHub notebook: jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

Circles and markers	**Added:** at each launch site **Why:** show where every site is on the map
Marker cluster	**Added:** green (landed) and red (failed) markers per launch **Why:** compare outcomes at each site at a glance
Mouse position	**Added:** shows coordinates under the cursor **Why:** find the coordinates of nearby features
Lines and distance markers	**Added:** from a site to the nearest coastline, railway, highway and city **Why:** measure how close sites are to these features

GitHub notebook: lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

Launch site dropdown

Added: Choose all sites or one site

Why: Filter both charts to the site you want to inspect

Success pie chart

Added: All sites: share of successful launches per site. One site: successes vs. failures

Why: Shows which sites contribute most of the successes

Payload range slider

Added: Pick a payload range in kg

Why: Check whether success changes with payload mass

Payload vs. outcome scatter

Added: Payload mass against outcome, colored by booster version

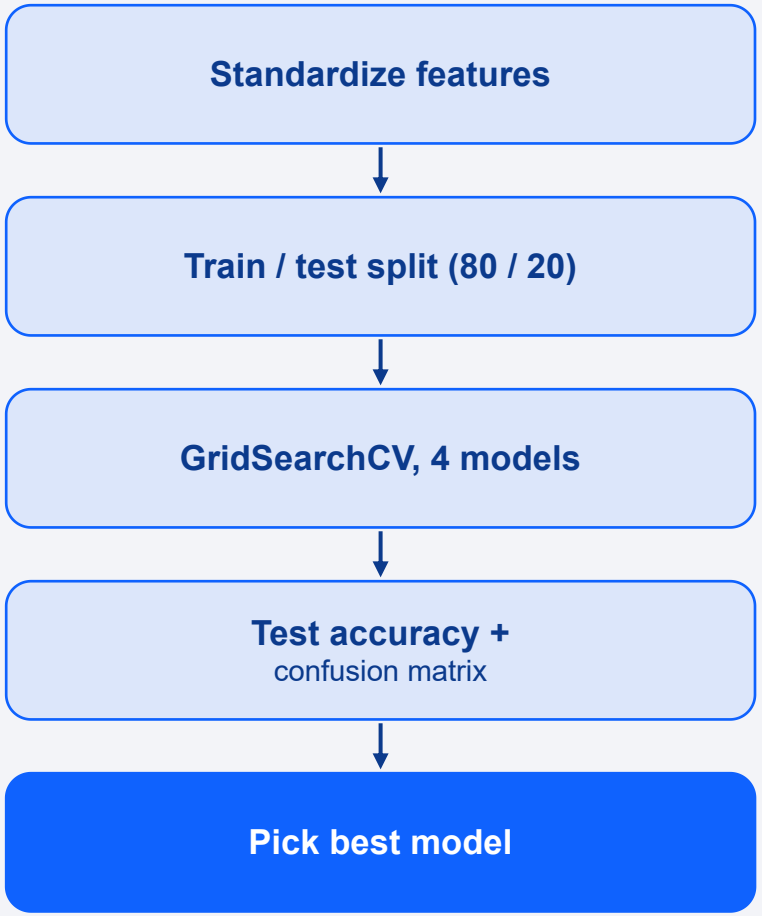
Why: Shows which payload ranges and boosters succeed most

GitHub file: spacex-dash-app.py

Predictive Analysis (Classification)

- **Prepare:** Class column as the target Y; one-hot encoded features as X, standardized with StandardScaler
- **Split:** train_test_split with 20% of the data held out for testing
- **Build and tune:** Logistic Regression, SVM, Decision Tree and KNN, each tuned with GridSearchCV using 10-fold cross-validation
- **Evaluate:** accuracy on the test set and a confusion matrix for each model
- **Select:** compare test accuracy and pick the best model

GitHub notebook:
SpaceX_Machine_Learning_Prediction_Part_5.ipynb



Results

Exploratory data analysis

66.7%

of first stages landed. Success rose from 0% (2010 to 2013) to about 90% (2019)

Interactive analytics

76.9%

success rate at KSC LC-39A, the best of the four launch sites

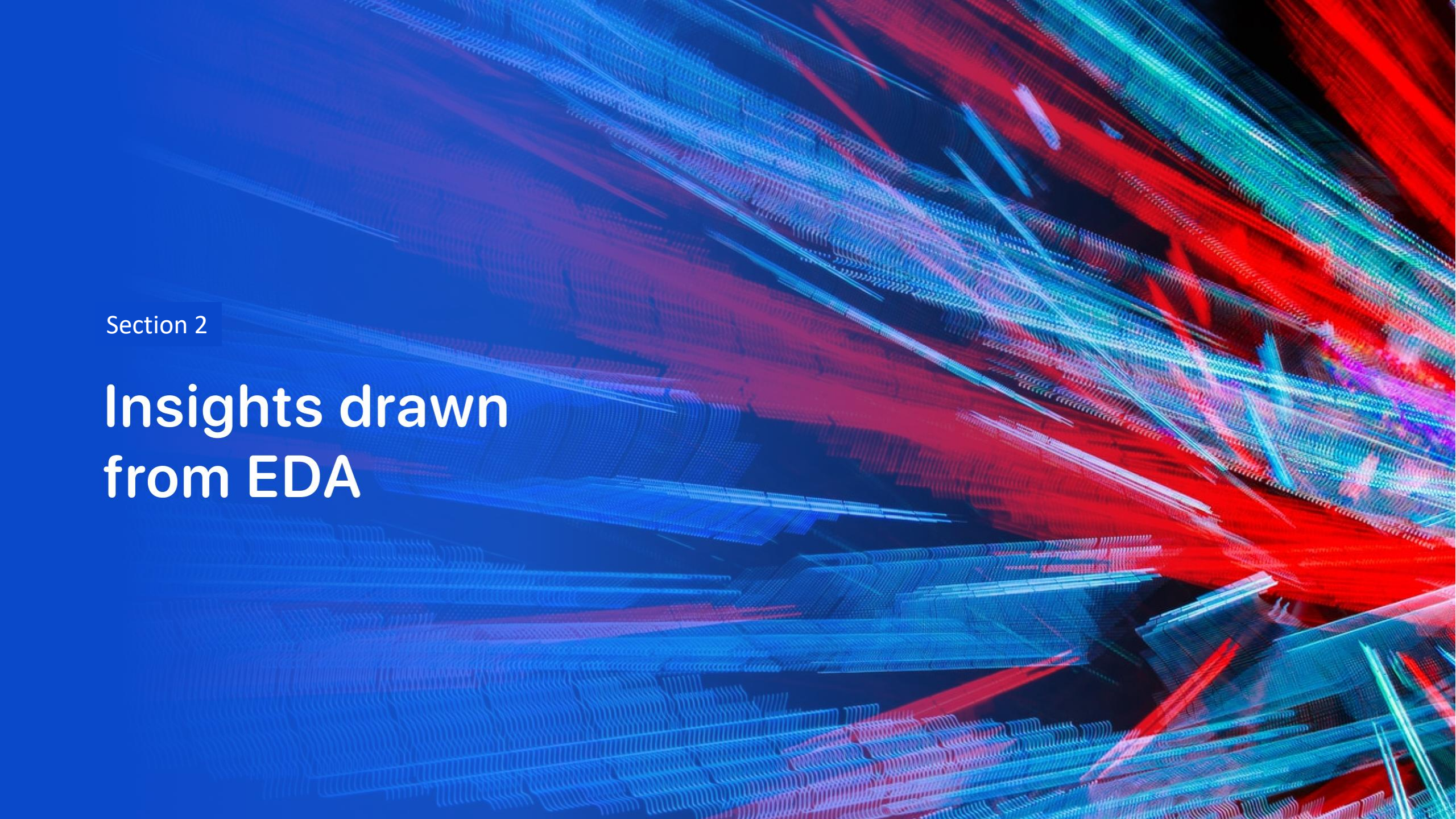
Predictive analysis

83.3%

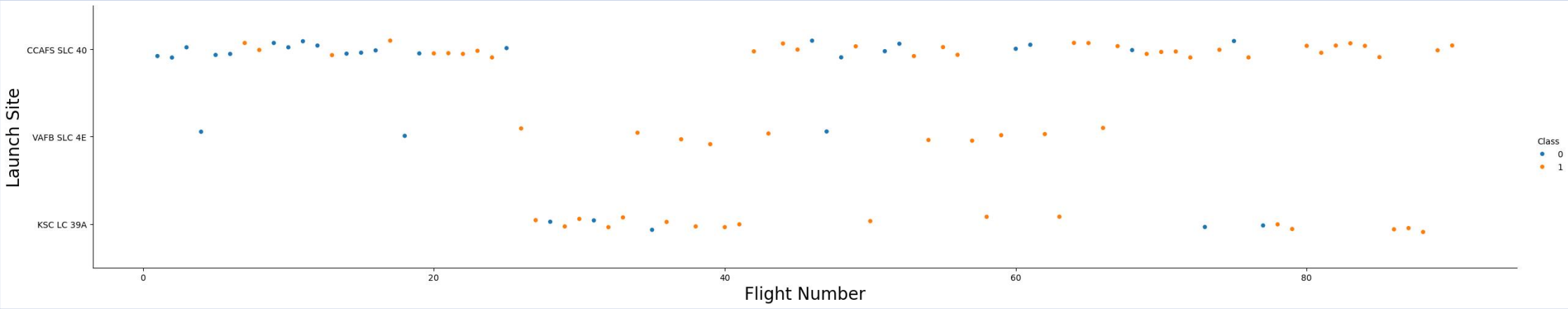
test accuracy for all four models. Decision Tree best in cross-validation (0.891)

Section 2

Insights drawn from EDA



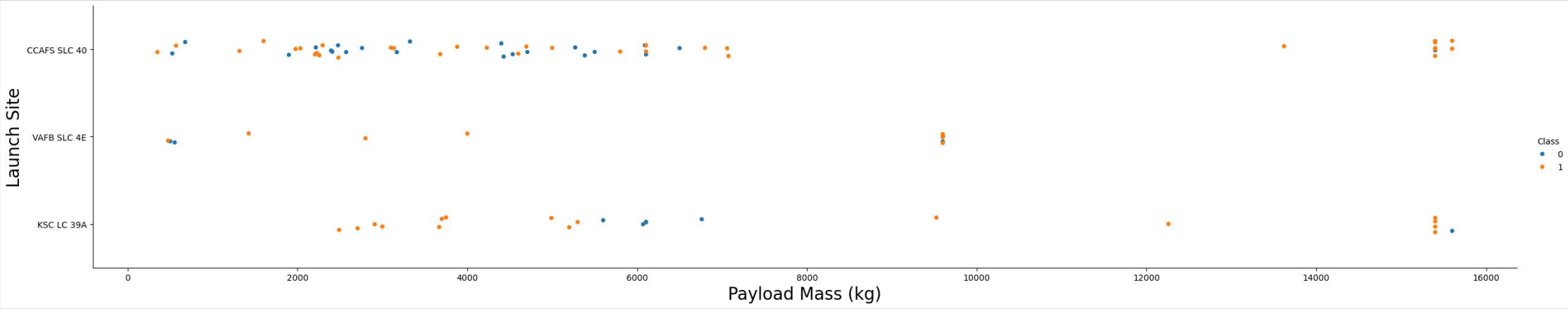
Flight Number vs. Launch Site



What the chart shows (orange = landed, blue = did not land)

- CCAFS SLC 40 hosts most launches, including nearly all of the early ones
- Early flights (roughly the first 20) mostly failed to land; later flights mostly landed at every site
- So experience (flight number) matters more for landing success than the site itself

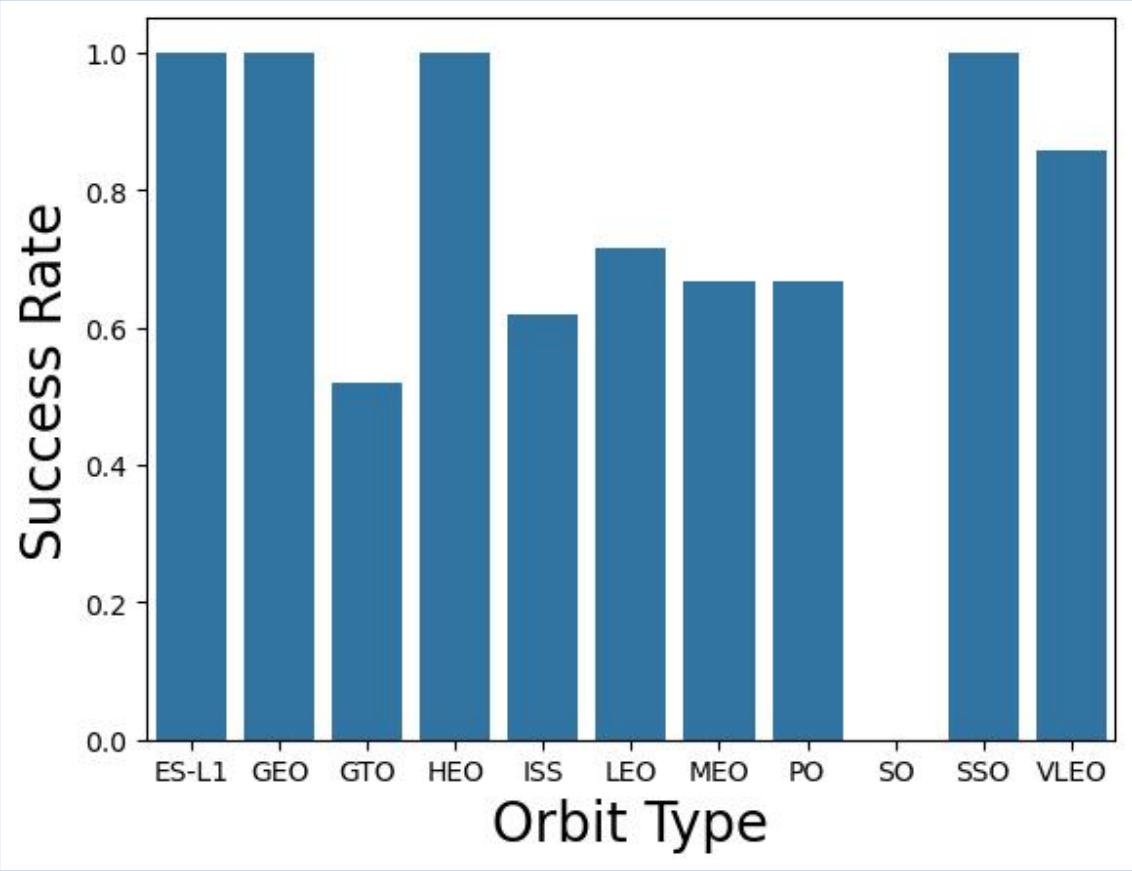
Payload vs. Launch Site



What the chart shows

- VAFB SLC 4E launched no payloads above 10,000 kg
- The heaviest payloads (about 13,000 to 16,000 kg) launched from CCAFS SLC 40 and KSC LC 39A, and nearly all of them landed
- Light and medium payloads show both outcomes at every site, so payload alone does not decide the outcome

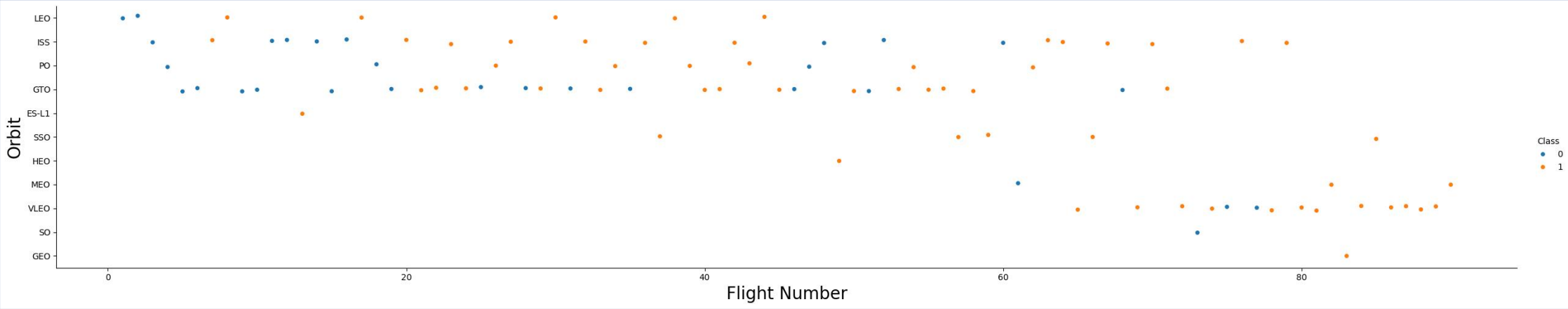
Success Rate vs. Orbit Type



What the chart shows

- ES-L1, GEO, HEO and SSO show a 100% success rate, and SO shows 0%
- But ES-L1, GEO, HEO and SO had only one launch each (SSO had 5), so those rates rest on very little data
- Among the most common orbits, VLEO (14 launches) is highest at about 86%, and GTO (27 launches) is lowest at about 52%

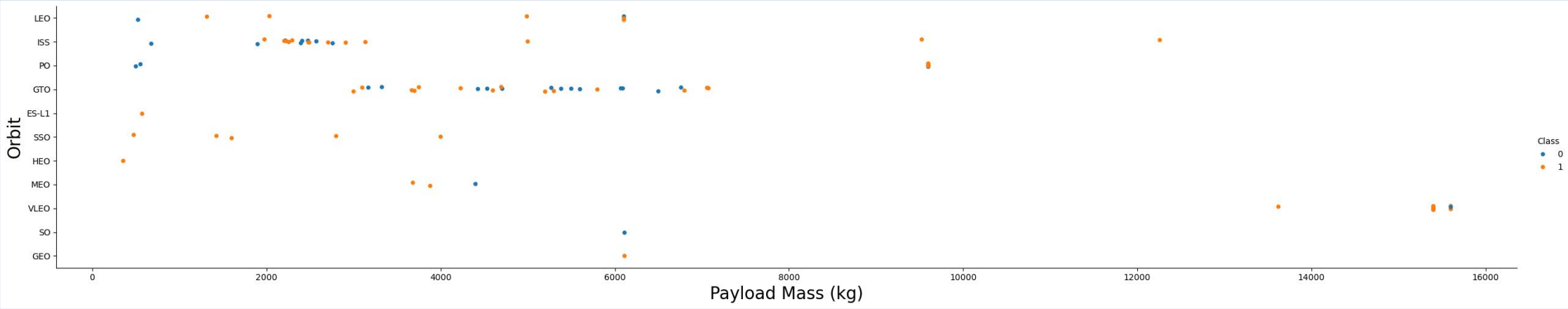
Flight Number vs. Orbit Type



What the chart shows

- In LEO, success seems to grow with flight number
- In GTO, there is no clear link between flight number and success
- VLEO only appears in later flights (after about flight 65), and most of those landed

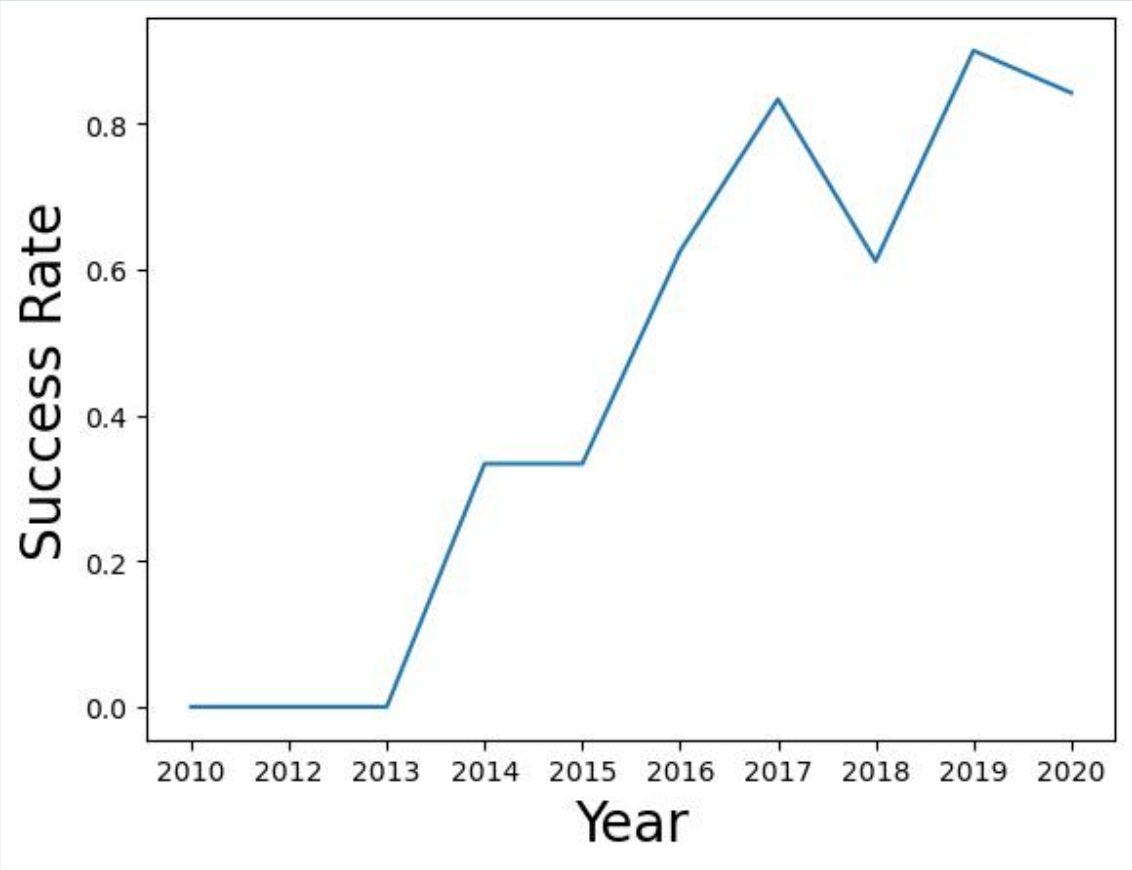
Payload vs. Orbit Type



What the chart shows

- With heavy payloads, landings succeed more often for Polar (PO), LEO and ISS orbits
- GTO missions mix successes and failures across the 3,000 to 7,000 kg range, so the two are hard to tell apart
- VLEO missions carry the heaviest payloads (about 13,600 to 15,600 kg)

Launch Success Yearly Trend



What the chart shows

- No first stage landed from 2010 to 2013
- The success rate climbed after 2013, reaching about 83% in 2017
- It dipped to about 61% in 2018, then peaked at about 90% in 2019
- Overall, landing success improved steadily as SpaceX gained experience

All Launch Site Names

```
SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Explanation: DISTINCT returns each value once, so the data has 4 unique launch sites: three in Florida (Cape Canaveral and Kennedy Space Center) and one in California (Vandenberg).

Launch Site Names Begin with 'CCA'

```
SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Date	Booster	Launch Site	Payload	Mass (kg)	Customer	Landing Outcome
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	Dragon Qualification Unit	0	SpaceX	Failure (parachute)
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1	0	NASA (COTS) NRO	Failure (parachute)
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	NASA (COTS)	No attempt
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	NASA (CRS)	No attempt
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	NASA (CRS)	No attempt

Explanation: LIKE 'CCA%' matches any site name starting with CCA. These first 5 records are the earliest Falcon 9 v1.0 flights from CCAFS LC-40, all with light payloads and no successful landing. Some columns are hidden to fit the slide.

Total Payload Mass

```
SELECT SUM(PAYLOAD_MASS__KG_) AS total_payload_mass
FROM SPACEXTABLE
WHERE Customer = 'NASA (CRS)';
```

total_payload_mass (kg)
45,596

Explanation: SUM adds up the payload mass of every launch where NASA's Commercial Resupply Services (CRS) was the customer: 45,596 kg in total.

Average Payload Mass by F9 v1.1

```
SELECT AVG(PAYLOAD_MASS__KG_) AS avg_payload_mass
FROM SPACEXTABLE
WHERE Booster_Version = 'F9 v1.1';
```

avg_payload_mass (kg)
2,928.4

Explanation: AVG gives the mean payload mass for launches using booster version F9 v1.1: about 2,928 kg.

First Successful Ground Landing Date

```
SELECT MIN(Date) AS first_ground_pad_success
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)';
```

first_ground_pad_success
2015-12-22

Explanation: MIN on the date column returns the earliest date, so the first successful ground pad landing was on 22 December 2015.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
SELECT Booster_Version FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
  AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Explanation: Four boosters landed on a drone ship while carrying between 4,000 and 6,000 kg. All are Falcon 9 FT (Full Thrust) boosters, two of them on reused cores (.2).

Total Number of Successful and Failure Mission Outcomes

```
SELECT Mission_Outcome, COUNT(*) AS total
FROM SPACEXTABLE
GROUP BY Mission_Outcome;
```

Mission_Outcome	total
Failure (in flight)	1
Success	98
Success (trailing space)	1
Success (payload status unclear)	1

Explanation: GROUP BY counts each outcome. 'Success' appears twice because one value has a trailing space, so there are really 99 plain successes, 1 success with unclear payload status, and only 1 in-flight failure out of 101 missions.

Boosters Carried Maximum Payload

```
SELECT DISTINCT Booster_Version FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ =
  (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

Booster_Version	Booster_Version	Booster_Version
F9 B5 B1048.4	F9 B5 B1051.4	F9 B5 B1051.6
F9 B5 B1049.4	F9 B5 B1049.5	F9 B5 B1060.3
F9 B5 B1051.3	F9 B5 B1060.2	F9 B5 B1049.7
F9 B5 B1056.4	F9 B5 B1058.3	
F9 B5 B1048.5		

Explanation: The subquery finds the largest payload mass, and the outer query lists every booster that carried it. All 12 are Block 5 (B5) boosters, most of them reused several times.

2015 Launch Records

```
SELECT substr(Date, 6, 2) AS month, Landing_Outcome,
       Booster_Version, Launch_Site
FROM SPACEXTABLE
WHERE substr(Date, 0, 5) = '2015' AND Landing_Outcome = 'Failure (drone ship)';
```

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Explanation: SQLite has no month names, so substr pulls the month and year out of the date text. In 2015 there were two failed drone ship landings, in January and April, both F9 v1.1 boosters from CCAFS LC-40.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
SELECT Landing_Outcome, COUNT(*) AS outcome_count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome ORDER BY outcome_count DESC;
```

Landing_Outcome	outcome_count	Landing_Outcome	outcome_count
No attempt	10	Controlled (ocean)	3
Success (drone ship)	5	Uncontrolled (ocean)	2
Failure (drone ship)	5	Failure (parachute)	2
Success (ground pad)	3	Precluded (drone ship)	1

Explanation: Ranked from most to least common (read the left pair first, then the right). In this early period, 'No attempt' was the most common outcome, and drone ship landings were split evenly: 5 successes and 5 failures.


Section 3
Launch Sites
Proximities Analysis

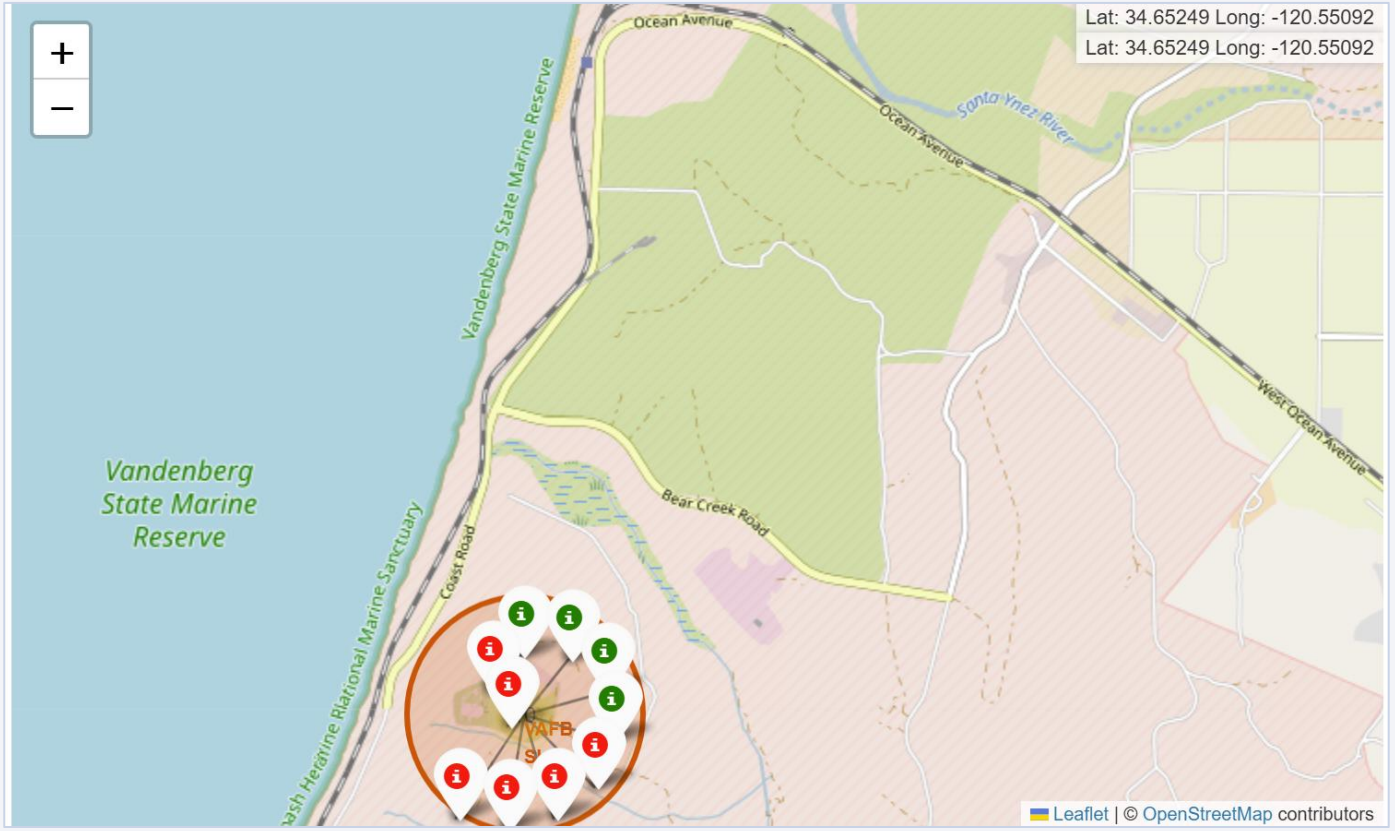
All Launch Sites on a Map



Findings

- There are 4 launch sites: 3 in Florida and 1 in California (VAFB SLC-4E)
- Every site is right on the coast: the Atlantic in Florida, the Pacific in California
- The three Florida sites are clustered together on Cape Canaveral, so their labels overlap at this zoom level
- All sites are in the southern US, which puts them closer to the Equator

Launch Outcomes by Site: VAFB SLC-4E

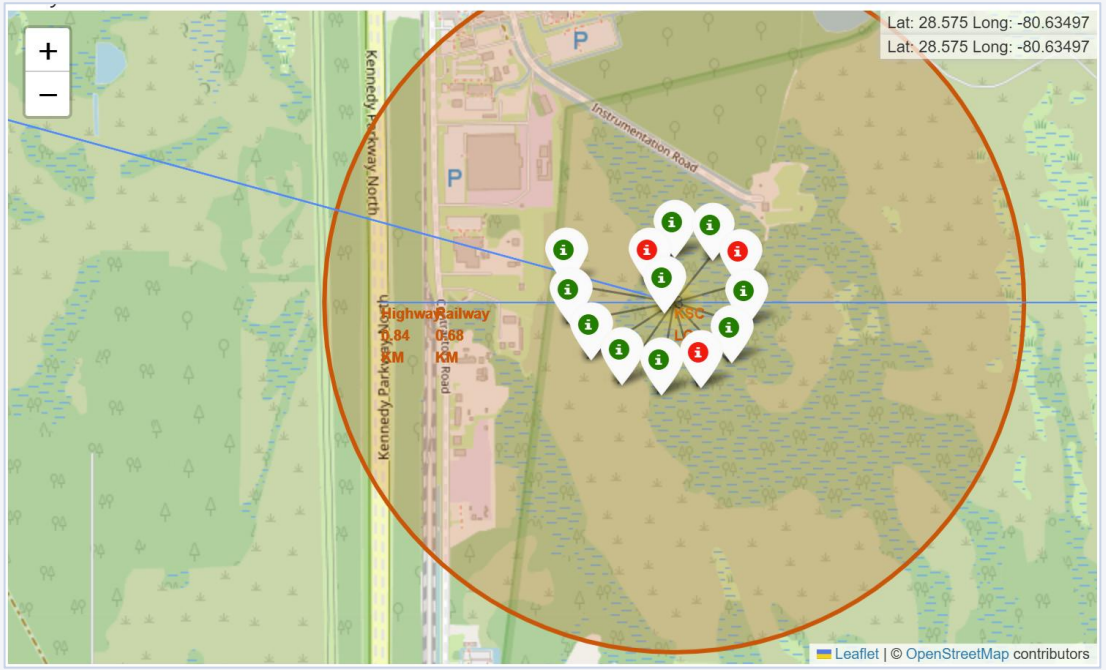


Findings

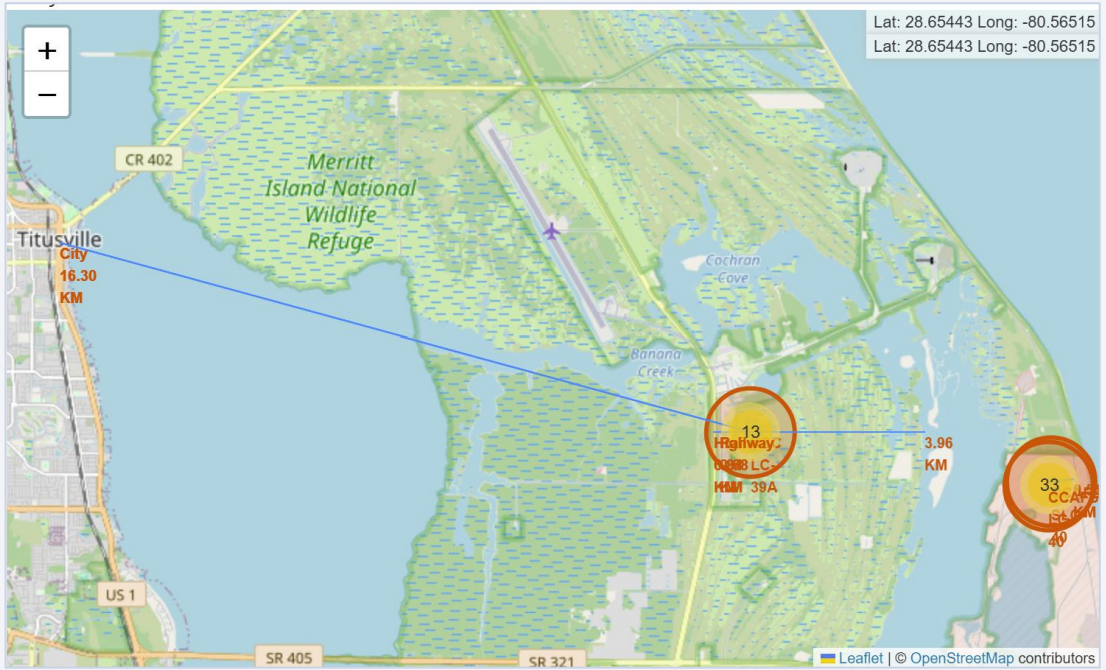
- Green pins are landings (class 1), red pins are failures (class 0)
- VAFB SLC-4E: 4 green and 6 red, so 4 of 10 launches (40%) landed
- Marker clusters group the launches at each site: 26 at CCAFS LC-40, 7 at CCAFS SLC-40, 13 at KSC LC-39A and 10 at VAFB SLC-4E (56 in total)

Distances from KSC LC-39A to Its Proximities

Close-up: railway and highway



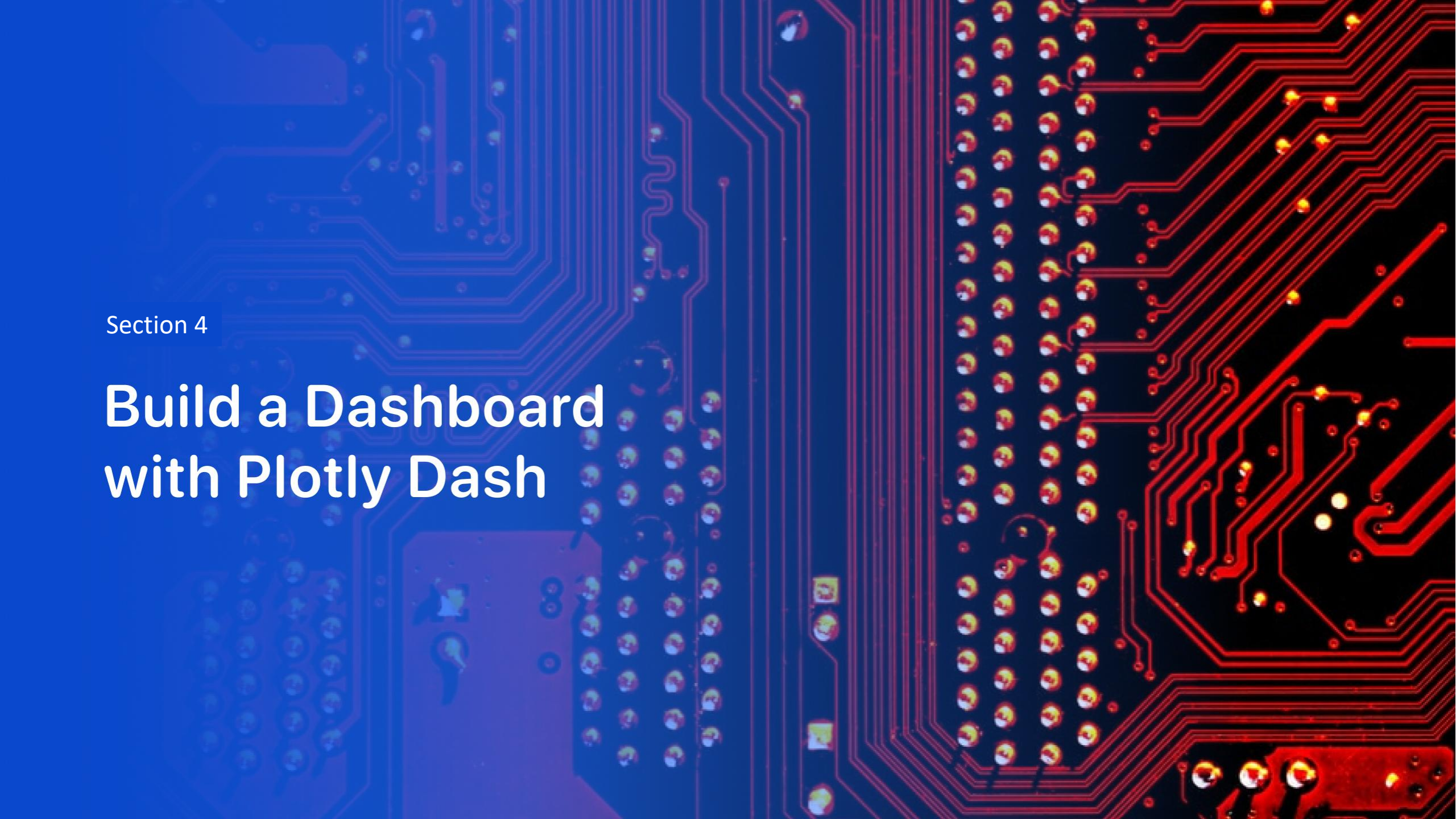
Wider view: coastline and city



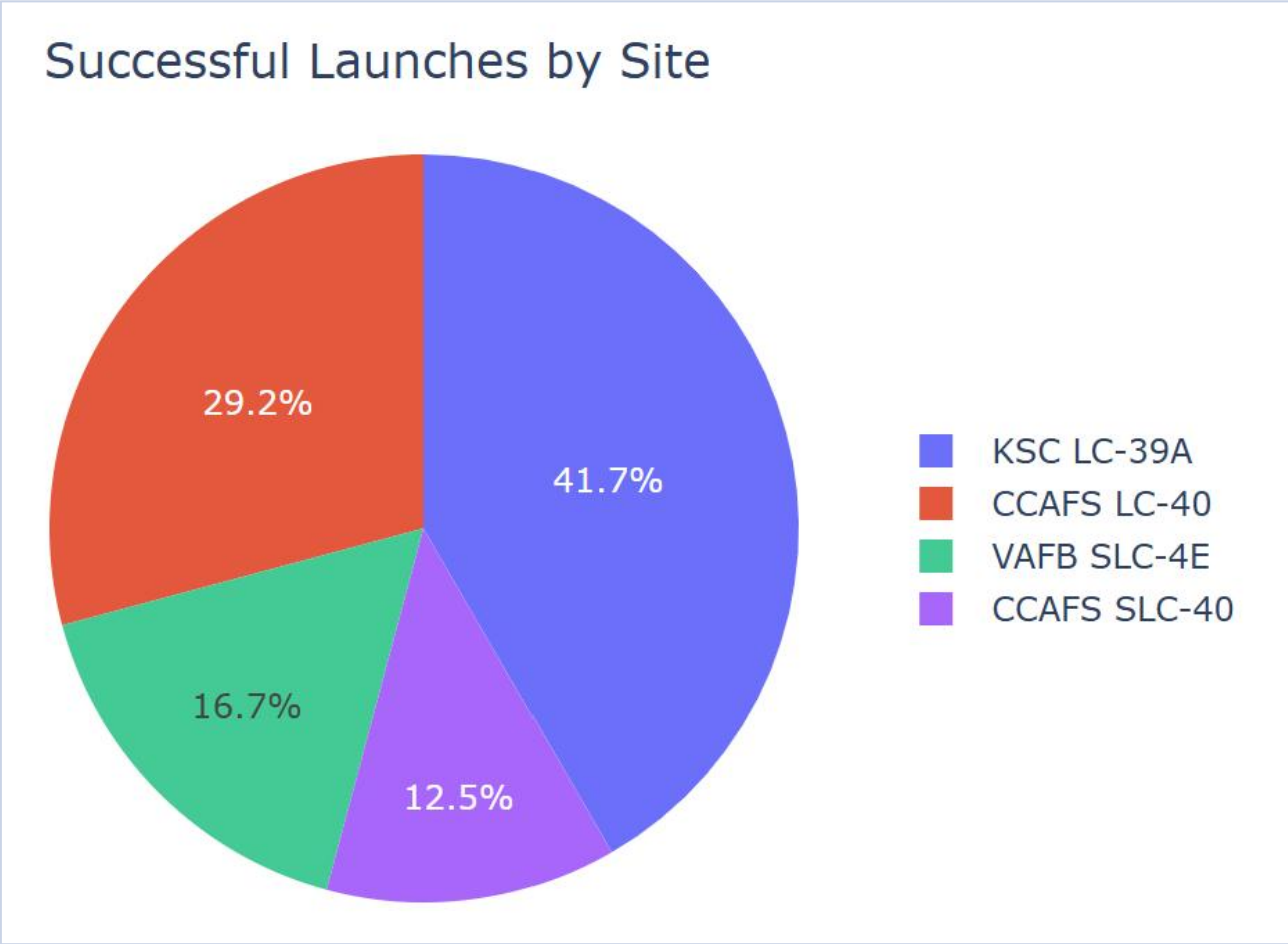
Proximity	Distance
Railway	0.68 km
Highway	0.84 km
Coastline	3.96 km
City (Titusville)	16.30 km

Findings

- Very close to a railway and highway, so rockets and equipment are easy to move
- Only about 4 km from the coast, so launches go out over the ocean
- Over 16 km from the nearest city, which keeps people at a safe distance


Section 4
Build a Dashboard with Plotly Dash

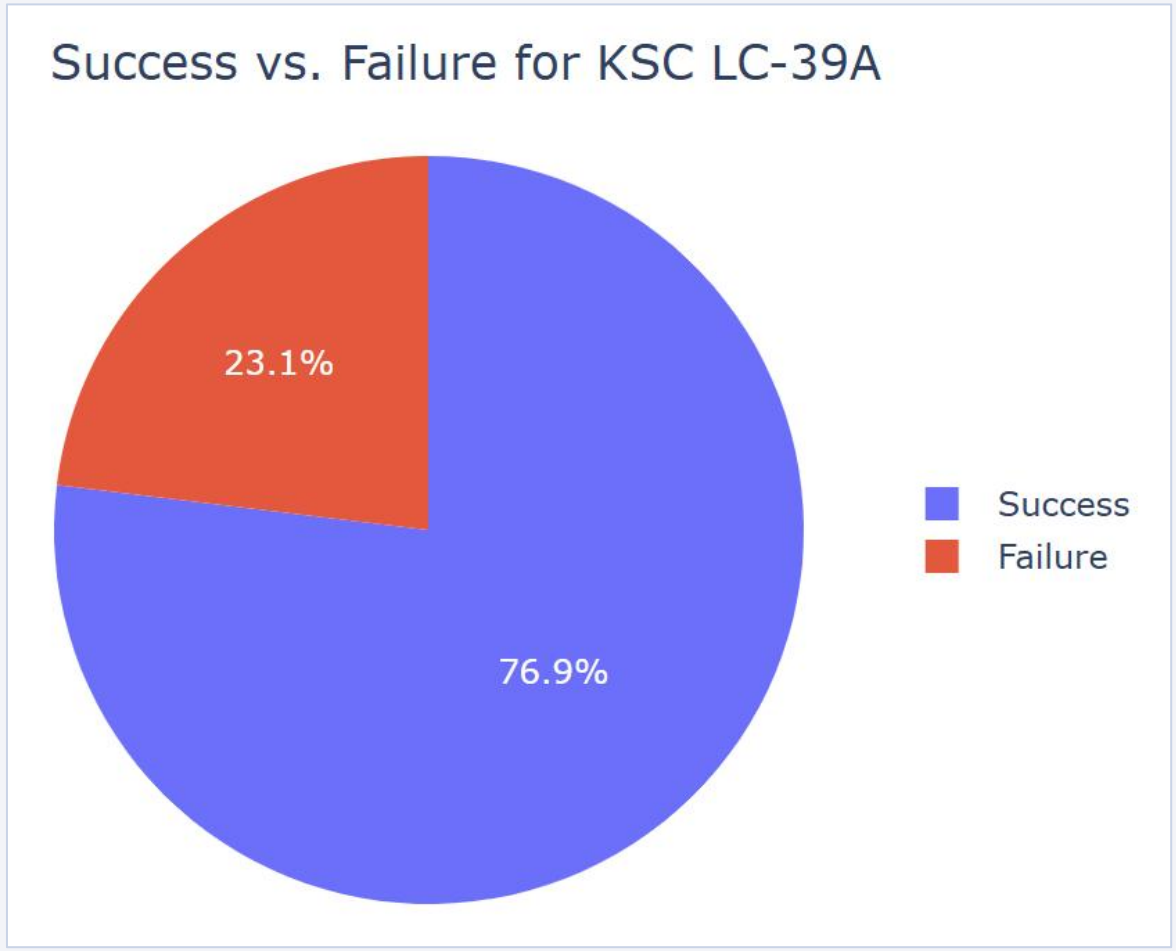
Successful Launches by Site



Findings

- KSC LC-39A has the largest share of all successful launches (41.7%)
- CCAFS LC-40 is second (29.2%), then VAFB SLC-4E (16.7%) and CCAFS SLC-40 (12.5%)
- This chart shows share of successes, not success rate: a site with more launches can have a bigger slice

Highest Success Rate: KSC LC-39A



Site	Success rate
KSC LC-39A	76.9% (10/13)
CCAFS SLC-40	42.9% (3/7)
VAFB SLC-4E	40.0% (4/10)
CCAFS LC-40	26.9% (7/26)

Findings

- KSC LC-39A has the highest success rate: 10 of its 13 launches landed (its Folium map shows 10 green and 3 red pins)
- CCAFS LC-40 had the most launches but the lowest success rate, because it hosted many of the early flights

Payload vs. Launch Outcome for All Sites

Payload 0 to 5,000 kg



Payload 5,000 to 10,000 kg



Findings (class 1 = landed, 0 = did not land)

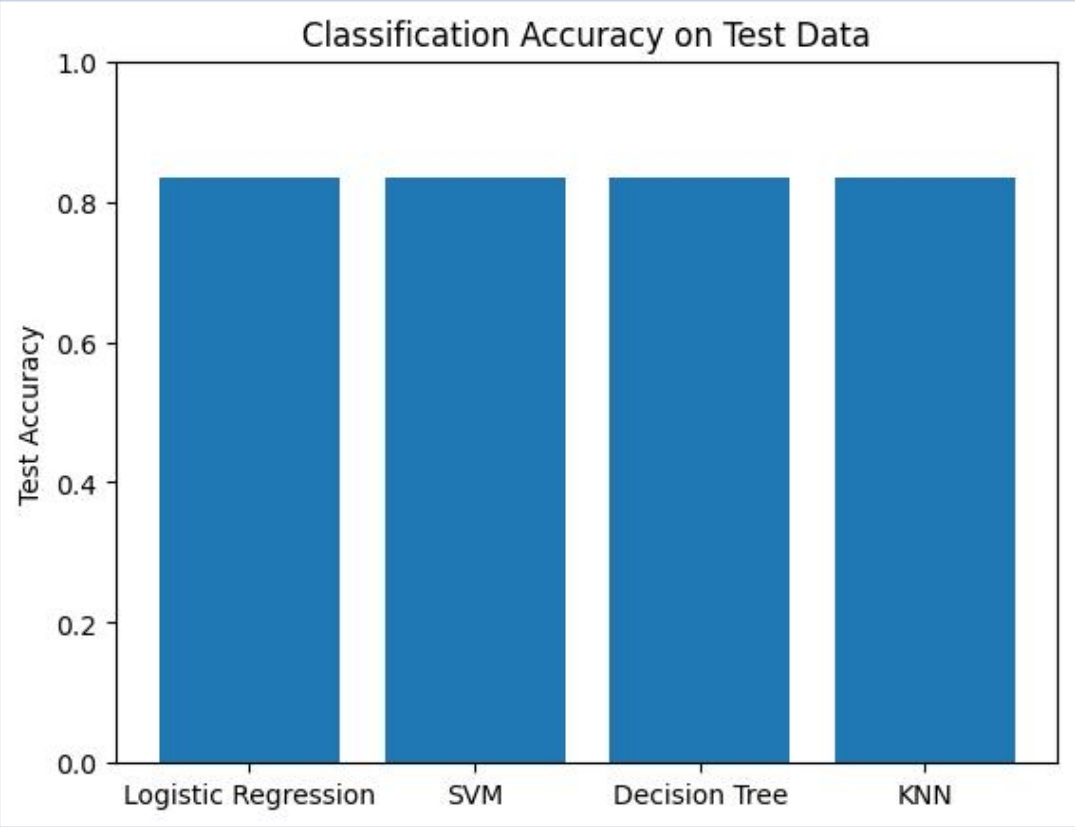
- Light to medium payloads (0 to 5,000 kg) have more successful landings than heavy ones
- From 5,000 to 10,000 kg, only 3 of 11 launches landed
- The FT booster has the most successes; v1.0 and v1.1 boosters mostly failed to land

Section 5

Predictive Analysis (Classification)



Classification Accuracy

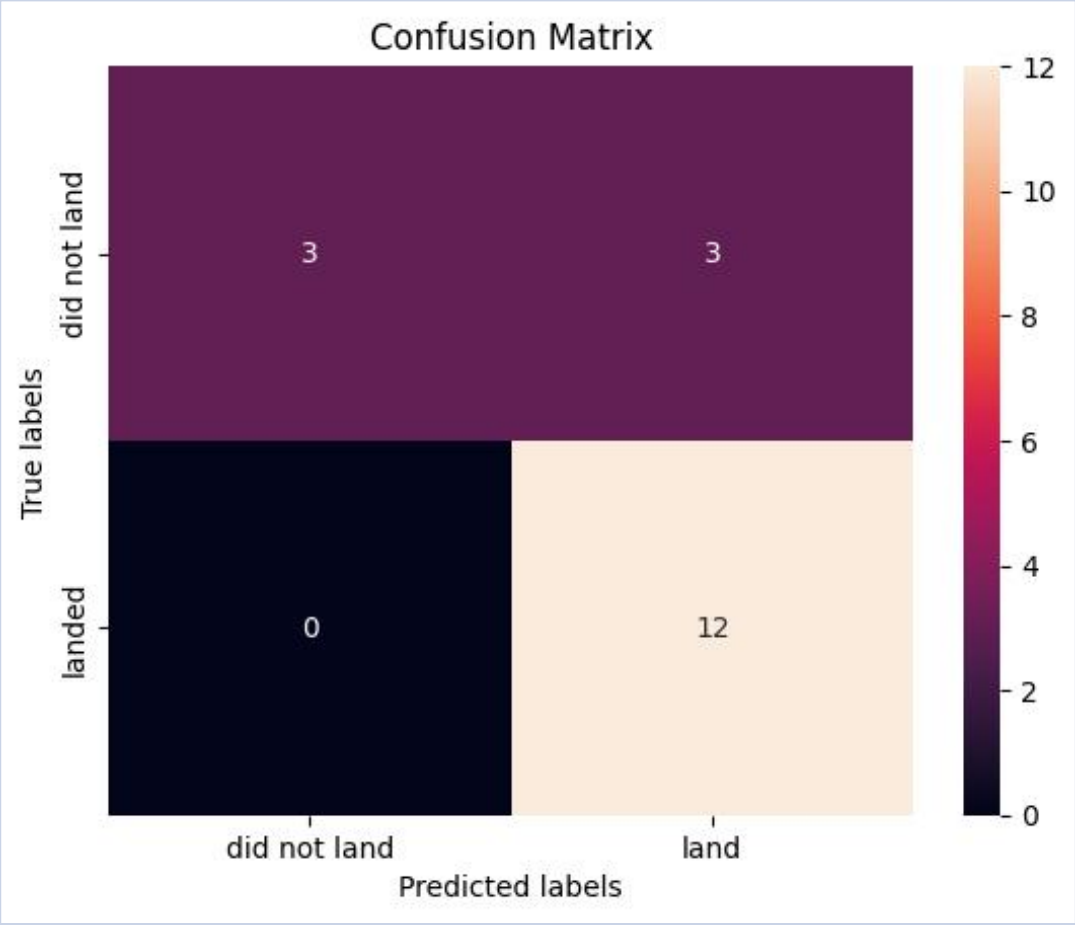


Model	Test accuracy	CV accuracy
Logistic Regression	0.833	0.846
SVM	0.833	0.848
Decision Tree	0.833	0.891
KNN	0.833	0.848

Which model is best?

- All four models scored the same test accuracy: 83.3%, or 15 of 18 test launches
- With a tie on the test set, cross-validation accuracy (10 folds on the training data) breaks it: the Decision Tree is best at 0.891
- The tree uses a random splitter with no fixed seed, so its score varies slightly between runs

Confusion Matrix



Decision Tree confusion matrix

- 12 true positives: landings predicted correctly
- 3 true negatives: failures predicted correctly
- 3 false positives: failures the model predicted as landings
- 0 false negatives: no landing was missed
- All four models produced this same matrix. Their weak point is false positives: they are too optimistic about landings

Conclusions

1

Experience matters

Landing success rose from 0% (2010 to 2013) to about 90% in 2019, and later flights land far more often.

2

Site matters

KSC LC-39A has the highest success rate (76.9%). All sites sit near coasts, railways and highways, away from cities.

3

Orbit and payload matter

GTO has the lowest success rate of the common orbits. Heavy payloads land well for PO, LEO and ISS orbits.

4

Landing is predictable

All four models predict landing with 83.3% test accuracy; the Decision Tree is best in cross-validation.

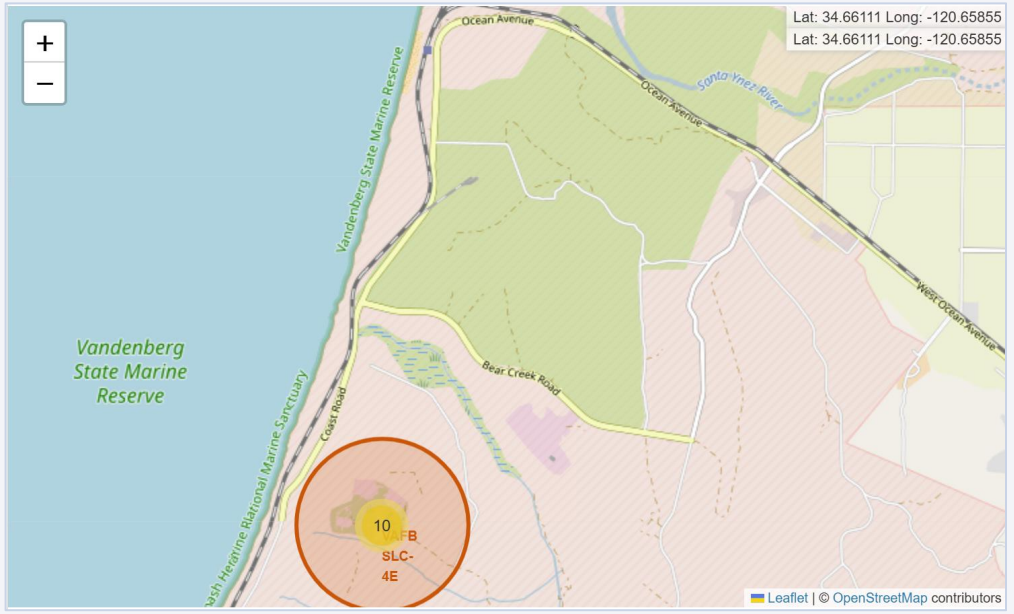
A competitor can use these models to estimate whether a first stage will be reused, and so estimate the cost of a SpaceX launch.

Appendix

Project files on GitHub

- jupyter-labs-spacex-data-collection-api.ipynb
- jupyter-labs-webscraping.ipynb
- labs-jupyter-spacex-Data_wrangling.ipynb
- jupyter-labs-eda-sql-coursera_sqllite.ipynb
- edadataviz.ipynb
- lab_jupyter_launch_site_location.ipynb
- spacex-dash-app.py
- SpaceX_Machine_Learning_Prediction_Part_5.ipynb

Extra: VAFB SLC-4E surroundings



The California site shows the same pattern as Florida: a railway runs along the shoreline and Coast Road passes right by the site.

Thank you!

